

Development and validation of a clinical wearable deep learning based continuous in-hospital deterioration prediction model

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This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Summary

Thank you for the opportunity to review this manuscript. I would like to commend the authors for their initiative in this study. I think this paper should be accepted for publication with some minor edits. Overall, this work constitutes a noteworthy effort that highlights the practical utility of integrating wearable technology with advanced machine learning algorithms with valid points raised by the authors throughout the paper.

Introduction

No major comments. Authors address well the healthcare context of their proposal of a deep learning algorithm based on continuous monitoring with the primary aim of early identification of clinical deterioration aimed at ward-based patients.

Methods

No sample size calculation has taken place. This sample size is not clearly explained or justified how it was reached to and if it was sufficient to answer the research question.

The authors do not describe the eligibility criteria for the study participants. Although this omission is acknowledged later in the discussion section as a limitation, the lack of clear criteria remains unexplained. This shortcoming results in a small number of events (hard outcomes), rendering the deep learning model training infeasible for this particular task. Furthermore, in a clinical setting, practical considerations for resource allocation would likely necessitate the use of continuous monitoring through wearable devices for patients at somewhat medium or high risk of clinical deterioration to rationalize their use and ensure cost-effectiveness.

Which modalities do the signals 1 and 2 belong to in the equation provided?

Authors outline the pre-processing techniques that were used to handle the wearable data along with the electronic records. A limitation with differences between the sampling rates of the devices that authors tried to address with linear interpolation. Missing data handling is well described with reasoning for data removal.

Please state clearly if the "MEWS > 6 for 30 minutes or more" was the ground truth that the model was developed on.

In the paragraph "Predictive deterioration modeling" it is unclear how the authors reached the 5.5 h sufficient data timeframe to forecast events of clinical deterioration. Can they please elaborate further?

Hard balance training mentioned in Figure 1 but not in the main body of the article. Is this used to tackle the class imbalance between the devices?

Results

Table 1: can the authors elaborate on the figures for each device group? This will give some context to the reader for the split/test validation process which is described later.

Figure 1- data from records referred as EMR but in text as EHR. Is this a typo?

Where SpO2 measurements included in the final model since the prospective device 1 does not provide this vital sign?

Why only RNN was chosen and only RNNs and Log Reg were tested, and no other ML model was attempted?

Discussion

Overall, well written addressing some important points. The authors report that respiration rate was the vital sign to account for the 71% of the alerts missed by the manual monitoring (EHR), and although it is briefly discussed that this might be related to the high frequency and richer nature of continuous monitoring, it does not explain why respiration rate is almost solely the most important feature. RR is generally subjectively measured, and recordings can be sometimes inconsistent. This finding is clinically significant. Can the authors explain a bit further what happens if CM is downsampled to match the frequency of manual recordings? How the model perform then?

The model is developed using vital signs and anthropometric features. No medical context is used or provided as to the nature of these patients (medical, surgical, severity of the condition that they are hospitalised for, prior physical status before admission).

These are clinically important factors that play important role in the frequency of vital signs measurement even outside ICU as healthcare staff are prioritising patients that are of higher risk and as a result the frequency of observation differs from the standard low risk patient. It is perhaps important to show/discuss the frequency of manual observations prior to the MEWS alert or the deterioration. Where all the deteriorations of sudden unexpected nature?

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors present interesting work where they develop a machine-learning based model to predict deterioration using data from two wearable devices. They aim to improve warning of rapid response team interventions, unplanned intensive care unit transfers, intubations, cardiac arrests, and in-hospital deaths. The stated goals were to assess whether these devices were a viable clinical alternative for monitoring on the medical wards, whether increased frequency of vital signs provide an advantage over episodic monitoring, and whether the data acquired combined with a machine learning approach could yield an accurate predictive algorithm.

I could not quite work out how the authors had decided to include particular patients or indeed hospitals and how the study had been powered. There does appear to be a study protocol but I could not access this. I am sorry but I found the paper hard to process, possibly partly because the aims are quite broad. One suggestion might be to follow a reporting guideline such as TRIPOD, though this only deals with part of the aims of the study.

The authors report the agreement between manual and wearable readings – noting that this is poor for respiratory rate. This agreement appears to be used both to decide what data to use for modelling and to assess whether the wearable devices are a viable alternative to manual measurements. They also report Retrospective and Prospective Device #1 and Alternate Device #2 temperature had 99% of all datasets' and devices' measurements fall within the 5% deviation line. This would be surprising for comparisons of skin surface and "Core" (assuming ear probe temperature measurement in the EHR – which is not reported), unless I have misunderstood what a 5% deviation line means. Again, unless I have misunderstood, the Bland-Altman plots seem to show that the 95% limits of agreement are above 1 degree celsius – which would be difficult in clinical practice.

I think the authors have then derived a model using a definition of instability of MEWS greater than 6 for ≥ 30 minutes – which I presume is derived from wearable data. They note that the wearable systems detect many more of these episodes than the EPR recordings – and that this is mainly due to differences in wearable recorded respiratory rate. I don't think its possible in this design to determine the usefulness or otherwise of these additional alerts.

They then test the model against hard outcomes. I cannot determine whether there were 17 in 17 different patients as suggested by table 1 or 11 as suggested in table 3 – it would seem unlikely that some of these outcomes didn't occur in the same patient – though possibly they may have used the first outcome that occurred in a patient? I was not entirely sure how the lack of SpO2 in prospective device 1 was dealt with in prospective evaluation

The small number of events (presumably across both devices) is a problem in assessing the usefulness of wearable devices but is problematic for validating a model. Perhaps it would be better to present the work as a feasibility study?

(Remarks on code availability)

I can't comment as I haven't seen the code and would not be expert to interpret it.

Reviewer #3

(Remarks to the Author)

The authors have conducted a thorough multi-validation investigation for a wearable based rnn tool for detecting early onset of inpatient deterioration. Overall, a good study but still some major comments that need addressing.

Some comments:

1. Please provide a table with what EHR data and what CM or wearable data provides in terms of variables. easier to refer to and explain
2. mews score is not clear to me. please describe in clearer mathematical language. how does the metric work - intuition is not clear either.
3. a figure on a hypothetical patient for prediction would be useful. i.e. here is patient values, at some point is "baseline" or start of follow-up and some point is the event deemed so by episodic monitoring (i believe?)
4. clearly describe what is the outcome being predicted in mews score section (ie., > 30-minute periods of elevated MEWS, ...)
5. I am a fan of the wearable data evaluation. as RR was inaccurately measured, could there be some scaling/calibrating that can make the the measurements fall within the 10% deviation line? that way you use as much data as possible...
6. please put the details on logistic regression and other benchmark model in the method section. what variables etc?
7. in alternative device, why do you think auroc is higher for log reg - ehr model?
8. this is quite odd. please include discussion material on this... and potentially text in limitation section as well.
9. also, device 1 prospective validation shows that ehr model LR has comparable AUC w respect to rnn model with wearables data - 0.89 v 0.9. only Device 1 retrospective validation shows promise for the proposed methods. given this, generalisability of method is not certain. please discuss.
10. not enough events in table 3 to make any call here... huge limitation that needs to be taken into account - would be better if removed..

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Authors have made a significant effort to address comments from all reviewers. I have no further concerns for this manuscript to be published.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors have made good progress with answering the comments of the reviewers.

I would suggest that the abstract is clarified to ensure the reader understands that patients were included as part of a quality initiative, and prediction models have been retrospectively built and validated on these data. It should also be clear that the authors experimented with different data periods for prediction to create best model performance.

I think the use of "prospective" could unintentionally mislead the reader in the context of work done on a retrospective dataset. I think a retrospective dataset has been split to allow model development with unseen data held out for testing. If this is the case I think it would be better if this was made clear throughout.

Given the small number of events - prediction of events is at best initial work as the other reviewers note. If they are to be included in the abstract the small number of events should be made clear (81% of ? 11 events). The abstract should also include the proportion of time when monitors were applied where sufficient data quality waa available for model training/application.

I remain unclear how temperature agreements with 95% confidence intervals above 1 degree would be useful in clinical practice and I think unless I have misunderstood this this limitation needs clarifying. Presumably as the patch was calibrated by the nurse the first ? number of readings were excluded as they would match as a result of calibration?

However, with these suggested clarifications the work is interesting and has been improved by the thoughtful responses to the contributions of the other reviewers. Its a lot of work and the authors are to be congratulated on a thoughtful approach to a taxing problem

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The authors of the "Beyond episodic early warning systems: a continuous clinical alert system for early detection of in-hospital deterioration" have done a remarkable job editing and revising the manuscript. Very thorough work.

They have responded to my reviewer 3 comments pretty comprehensively. I appreciate all the work they have done adding figures/tables and text that help explain the prediction objective, baseline, cohort, etc. However I still have some concerns on the utility and impact of the proposed model.

Comments:

1. "The patients that were patched were selected based often on the best clinical judgement of the staff in each of the wards where the wearable devices were deployed. In practice, this meant the nurse overseeing the patient would often decide whether to have a wearable device placed on the patient or not based on their assessment of how at risk the patient was for decompensating." This means that the selection of patients is subject to selection biases. What were the conditions they were chosen? In a really troubling hypothetical setting, patients were chosen in such a way that would "benefit" the model?
2. Also how generalizable is the model to the general population of patients for which you want to put the patch on and the wearables device? Yes indeed the model has been validated not on a different cohort but an alternate device. This is not the same as cohort being different. So as there were particular selection protocols for this cohort under Northwell's approval, how do we know this model is generalizable across cohorts... this is worth mentioning at least? How different are generally patients for this? What does this mean for the modelling? is it for understanding deterioration better as it is likely that it cannot be used for predictive modelling in the generalisable sense?
3. "Class imbalance was addressed by first randomly removing samples from the negative majority class and oversampling the positive class until we achieve a balance of 95%/5% negative/positive classes of the training set, instead of the original 99.7%/0.3% class balance. " this could be really dangerous for calibration. Please check: <https://arxiv.org/html/2404.19494v1>
4. Given this, I recommend tests of calibration. Please conduct.
5. I like the paragraph on how the RNN v LR is similar for AUROC but not for AUPRC. Great.

(Remarks on code availability)

I believe no code available.

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have clarified the areas I suggested and have also made changes as suggested by the other reviewers. I believe the article is now clear for readers.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)

The answers to most of my previous comments were fine. However, the responses to the ones on calibration were far from acceptable in my eyes. Calibration is key for showing reliability of modelling.

1. Indeed you have a binary output, but this binary output is derived from continuous estimates. Otherwise you could not present confusion matrices and area under PRC - the latter for sure can be estimated as precision-recall estimates across the spectrum of risk threshold. So if you can't get risk estimates, then I ask how did you derive area under PRC? In other words, it is clear there is a continuous risk prediction happening here and there is binary output calculation following continuous risk estimation based on threshold - e.g., >0.8 is 1 and ≤ 0.8 is 0

2. While resampling may work on select cases without affecting calibration as you have shown in the references, the best way to show this point indeed is to show the calibration. So again, I urge you please show calibration curves.

(Remarks on code availability)

Did not as waiting for calibration results.

Version 3:

Reviewer comments:

Reviewer #3

(Remarks to the Author)

Excellent work doing full calibration analyses. Indeed, these are not optimal calibration results, but the transparency in reporting this in results and extended discussion sections is duly appreciated.

Some more points:

1. I don't agree with this statement: " At the same time, this establishes that this is not a risk stratification model, but rather a binary classification model that produces an actionable output, flagging patients for ICU consults and possible elevation of care. Finally, recalibration can correct model calibration across the observed ranges of patient risk." When you do binary classification indeed this stratification of risk. Those below the threshold are low risk and those above are high-risk. Hence not sure this is accurate to put in. Please remove.

2. Great to see recalibration works well and the model looks better calibrated throughout 3 analyses. Has table 2 been changed accordingly? recalibration would have effected the true/neg capture. Not necessary to change but good to note that if table 2 is based off pre- recalibrated estimates, that this is written in footnotes of table next to abbrev? Also for device 1 prospective and alternate device #2 does this mean the threshold of interest is reset? there are no predictions above 0.1 for alternate device #2 and hence the threshold of 0.9 would not make much sense now?

3. Also the precision-recall curve would have changed as well. So any notes or thoughts on this?

With recalibration, I would say all discrimination metrics and analyses that rely on decision thresholds must be redone. While AUROC will remain the same generally, all others would move around. Recalibration is an important step and it is great to see it done. However, we can't see the full picture without all other analyses conducted given recalibrated estimates.

(Remarks on code availability)

Version 4:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The article is much improved by the changes made. I do not want to make significant additional work given the sequence of reviews that have been gone through. However, for the authors to consider.

1. I think it would be helpful to have flow diagrams to show how patients and data were included and excluded so that it is clear for example how there were 17 patients who deteriorated in table 1 but only 11 who are included in the assessment of performance.

I think it would be useful to the reader to qualify what the acceptable deviation lines for each vital were - so perhaps put in brackets next to 5/10% what these were - for temperature +/- 1-1.5 degree C across the relevant range - for example.

(Remarks on code availability)

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We wish to thank the reviewers for their thoughtful comments and suggestions. As you will see in this document, as well as the revised manuscript, the specific points raised by the reviewers have all been addressed; many clarifications have been added by the addition of one supplementary figure, two tables and many edits across all sections of the manuscript. We believe all this additional work has significantly improved the paper and we hope the revisions are satisfactory to the reviewers and the editors. Our responses are highlighted in blue and the corresponding additions/changes to the revised manuscript are highlighted in red. All line numbers refer to the red-lined manuscript.

Reviewer #1 (Remarks to the Author):

Summary

Thank you for the opportunity to review this manuscript. I would like to commend the authors for their initiative in this study. I think this paper should be accepted for publication with some minor edits. Overall, this work constitutes a noteworthy effort that highlights the practical utility of integrating wearable technology with advanced machine learning algorithms with valid points raised by the authors throughout the paper.

Introduction

No major comments. Authors address well the healthcare context of their proposal of a deep learning algorithm based on continuous monitoring with the primary aim of early identification of clinical deterioration aimed at ward-based patients.

Methods

No sample size calculation has taken place. This sample size is not clearly explained or justified how it was reached to and if it was sufficient to answer the research question.

- Both reviewer #1 and #2 are correct to point out that no sample size calculation was provided. We have corrected this oversight by adding sample size considerations for both our wearable data validation efforts, as well as the deep learning model convergence in the Supplementary section "Sample size estimation for wearable data validation and model convergence." We have added the following justification at lines 731-760:

Supplementary Table 4 Sample size estimation for wearable data validation and model convergence

To estimate the sample size necessary for validating the wearable data in this study, we calculated Cohen's d statistic using the following formula:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{SD}$$

Where $\bar{X}_1$ is the mean of the distribution of vital values measured from the wearable device and $\bar{X}_2$ is the mean of the distribution of vital values measured in the EHR. SD represents the expected standard deviation of the grouped data. Using this formula, we set out to test whether the difference in the means of the wearable and EHR vital values exceeded 10%, in other words, whether there was a 10% error or greater. To give a concrete example, assuming an expected average heart rate of 75bpm, a 10% difference in the means of the heart rates, from wearable and EHR sources, would translate to a difference of the means of at least 7.5bpm. The expected aggregate standard deviation of the heart rate from both sources is approximately 10 bpm. Entering these values into the equation above results in a Cohen's d of 0.75. For this study, we chose a power level of 95% with alpha = 0.01. Using these parameters and the following sample size lookup table:⁶²

$\alpha_2 = .01 (\alpha_1 = .005)$											
d											
Power	.10	.20	.30	.40	.50	.60	.70	.80	1.00	1.20	1.40
.25	725	183	82	47	31	22	17	13	9	7	6
.50	1329	333	149	85	55	39	29	22	15	11	9
.60	1603	402	180	102	66	46	34	27	18	13	10
2/3	1810	454	203	115	74	52	39	30	20	14	11
.70	1924	482	215	122	79	55	41	32	21	15	12
.75	2108	528	236	134	86	60	45	35	23	17	13
.80	2338	586	259	148	95	67	49	38	25	18	14
.85	2611	654	292	165	106	74	55	43	28	20	15
.90	2978	746	332	188	120	84	62	48	31	22	17
.95	3564	892	398	224	144	101	74	57	37	26	20
.99	4808	1203	536	302	194	136	100	77	50	35	26

Resulted in an estimated required sample size between 57 and 74. Since each of our participants had many measurements, our study far exceeded the required sample size to detect the error rate in vital sign measurement between the wearable device and EHR.

In addition to estimating the sample size for validating the wearable data, we also estimated the amount of data necessary for neural network convergence. The choice of cohort size was made based on other clinical studies with similar architectures,^{59, 60} our prior studies,⁶¹ as well as studies on VC dimensions of neural networks.⁵⁸ With this context in mind, we observed that LSTM RNNs typically converge on sample sizes with over 600 patients. Recruiting 888 patients was based on this consideration and the availability of patches during this study.

The authors do not describe the eligibility criteria for the study participants. Although this

omission is acknowledged later in the discussion section as a limitation, the lack of clear criteria remains unexplained. This shortcoming results in a small number of events (hard outcomes), rendering the deep learning model training infeasible for this particular task. Furthermore, in a clinical setting, practical considerations for resource allocation would likely necessitate the use of continuous monitoring through wearable devices for patients at somewhat medium or high risk of clinical deterioration to rationalize their use and ensure cost-effectiveness.

- Thank you for pointing out this omission. We have included the following paragraph in the “Data Source, patient population and eligibility criteria” section to address this omission (Line 131-137, Paragraph 1):

While our retrospective analysis was approved by IRB as a research study, the patching of patients was performed as a quality initiative, that augmented the standard of care for these patients. The patients that were patched were selected based often on the best clinical judgement of the staff in each of the wards where the wearable devices were deployed. In practice, this meant the nurse overseeing the patient would often decide whether to have a wearable device placed on the patient or not based on their assessment of how at risk the patient was for decompensating.

Which modalities do the signals 1 and 2 belong to in the equation provided?

- The inline equation for the Bland Altman has been modified to label signal 1 and 2 as follows (Line 167, Paragraph 5):

$$(\Delta = \text{Signal 1 (Wearable)} - \text{Signal 2 (EHR)})$$

Authors outline the pre-processing techniques that were used to handle the wearable data along with the electronic records. A limitation with differences between the sampling rates of the devices that authors tried to address with linear interpolation. Missing data handling is a well described with reasoning for data removal.

Please state clearly if the “MEWS > 6 for 30 minutes or more” was the ground truth that the model was developed on.

We have added the following paragraph to state this clearly in the section “Predictive deterioration modeling; Model training and testing ” (Line 242-247, Paragraph 3):

We trained a recurrent neural network (RNN) with long-short-term-memory (LSTM) units model that incorporated age, BMI, SBP, and continuously monitored vital signs (HR, RR, temperature, SpO2, MEWS value, and motion signals) to predict in-hospital deterioration with a 0.5-24 hour time horizon using a 5 hour sequence (Figure 2). We constructed the clinical alert labels detailed previously, based on smoothed MEWS values using a 30-minute-wide rolling average window.

In the paragraph “Predictive deterioration modeling” it is unclear how the authors reached the 5.5 h sufficient data timeframe to forecast events of clinical deterioration. Can they please elaborate further?

- Thank you for pointing out this omission. We have added the following line in the section “Predictive deterioration modeling; Artifact rejection, signal processing and model selection” (Lines 221-224, Paragraph 1):

In order to determine the optimal window length for predicting clinical alerts 0.5 to 24 hours in the future, we performed a grid search of window lengths from 1 to 12 hours at 1-hour increments and determined 5 hours to be the optimal length based on retrospective model performance.

Hard balance training mentioned in Figure 1 but not in the main body of the article. Is this used to tackle the class imbalance between the devices?

- Yes, we've added details on our class balancing to the section "Predictive deterioration modeling; *Model training and testing*" (Lines 254-256, Paragraph 5):

Class imbalance was addressed by first randomly removing samples from the negative majority class and oversampling the positive class until we achieve a balance of 95%/5% negative/positive classes of the training set, instead of the original 99.7%/0.3% class balance.

Results

Table 1: can the authors elaborate on the figures for each device group? This will give some context to the reader for the split/test validation process which is described later (Line 305):

- Table 1 has been modified to include three additional columns that breakout the demographic summaries for the "Retrospective, Alternative Device and Prospective" datasets.

Figure 1- data from records referred as EMR but in text as EHR. Is this a typo?

- Thank you for pointing this out. The EHR references in Figure 1 have been corrected.

Where SpO2 measurements included in the final model since the prospective device 1 does not provide this vital sign?

- SpO2 was still included in the final model and the SpO2 variable was imputed with all zeroes. This has been indicated in the section "Predictive deterioration modeling; *Artifact rejection, signal processing and model selection*" (Lines 238-239, Paragraph 2):

For the prospective device 1, SpO2 was not available so the variables was imputed with all zeroes for the final model.

Why only RNN was chosen and only RNNs and Log Reg were tested, and no other ML model was attempted?

- Based on our review of the literature, RNNs typically outperform most other modelling approaches with respect to continuous monitoring data. We added a line justifying our choice of model to the introduction (Lines 81-83, Paragraph 3):

Further, we and others have shown that deep learning models achieve the best performance on continuous monitoring data compared to many existing modelling approaches, hence our selection of a recurrent neural network, deep learning model^{57,61}.

Discussion

Overall, well written addressing some important points. The authors report that respiration rate was the vital sign to account for the 71% of the alerts missed by the manual monitoring (EHR), and although it is briefly discussed that this might be related to the high frequency and richer nature of continuous monitoring, it does not explain why respiration rate is almost solely the most important feature. RR is generally subjectively measured, and recordings can be sometimes inconsistent. This finding is clinically significant. Can the authors explain a bit further what happens if CM is downsampled to match the frequency of manual recordings? How the model perform then?

- We'd like to thank the reviewer for making this astute observation. To look at this we performed a down sampling analysis (Supplementary Figure 3) where we looked at the model performance as we progressively down sampled the continuous monitoring signal from an inter-sample interval of 1minute up to 300 minutes, performing a forward-fill imputation to fill values removed by down sampling. We'd like to point the reviewer to the Methods section titled "Lead time and sampling rate analysis" (Lines 427-448).

The model is developed using vital signs and anthropometric features. No medical context is used or provided as to the nature of these patients (medical, surgical, severity of the condition that they are hospitalized for, prior physical status before admission).

These are clinically important factors that play important role in the frequency of vital signs measurement even outside ICU as healthcare staff are prioritizing patients that are of higher risk and as a result the frequency of observation differs from the standard low risk patient. It is perhaps important to show/discuss the frequency of manual observations prior to the MEWS alert or the deterioration. Where all the deteriorations of sudden unexpected nature?

- We've added the following paragraph to the discussion addressing this point (Line 464-473, Paragraph 3):

The current standard of care in patients admitted on medical and surgical floors, where the study took place, implements a vital sign check every 3-4 hours depending on the hospital and unit where a patient is situated. When a patient reaches a MEWS score above seven, vital sign frequency is then increased to every 2 hours. A MEWS score of eight results in increased vital sign frequency and an independent provider evaluation. Supplementary Table 3 details the additional steps that are taken as the MEWS score increases beyond eight. In this study, we only looked at MEWS alerts (MEWS > 6 for 30 minutes or longer) that were not preceded by any other MEWS alerts for at least 24 hours. Therefore, the additional, earlier clinical alerts that were found in this study, represent truly unexpected deteriorations enabled by more frequent monitoring of the patient.

Reviewer #2 (Remarks to the Author):

The authors present interesting work where they develop a machine-learning based model to predict deterioration using data from two wearable devices. They aim to improve warning of rapid response team interventions, unplanned intensive care unit transfers, intubations, cardiac

arrests, and in-hospital deaths. The stated goals were to assess whether these devices were a viable clinical alternative for monitoring on the medical wards, whether increased frequency of vital signs provide an advantage over episodic monitoring, and whether the data acquired combined with a machine learning approach could yield an accurate predictive algorithm.

I could not quite work out how the authors had decided to include particular patients or indeed hospitals and how the study had been powered.

- Thank you for pointing out this omission. Reviewer #1 had similar remarks and we have included the following verbiage in the “Data Source, patient population and eligibility criteria” section to address this omission (Lines 131-137, Paragraph 1):

While our retrospective analysis was approved by IRB as a research study, the patching of patients was performed as a quality initiative, that augmented the standard of care for these patients. The patients that were patched were selected based often on the best clinical judgement of the staff in each of the wards where the wearable devices were deployed. In practice, this meant the nurse overseeing the patient would often decide whether to have a wearable device placed on the patient or not based on their assessment of how at risk the patient was for decompensating.

- In addition both reviewers #1 and #2 are correct to point out that no power justification was provided. We have corrected this oversight by adding a sample size justification in the supplementary section “Supplementary Table 4 .” We have added the following justification at (Lines 731-760):

Supplementary Table 4 Sample size estimation for wearable data validation and model convergence

To estimate the sample size necessary for validating the wearable data in this study, we calculated Cohen’s d statistic using the following formula:

$$d = \frac{\bar{X}_1 - \bar{X}_2}{SD}$$

Where $\bar{X}_1$ is the mean of the distribution of vital values measured from the wearable device and $\bar{X}_2$ is the mean of the distribution of vital values measured in the EHR. SD represents the expected standard deviation of the grouped data. Using this formula, we set out to test whether the difference in the means of the wearable and EHR vital values exceeded 10%, in other words, whether there was a 10% error or greater. To give a concrete example, assuming an expected average heart rate of 75bpm, a 10% difference in the means of the heart rates, from wearable and EHR sources, would translate to a difference of the means of at least 7.5bpm. The expected aggregate standard deviation of the heart rate from both sources is approximately 10 bpm.

Entering these values into the equation above results in a Cohen's d of 0.75. For this study, we chose a power level of 95% with alpha = 0.01. Using these parameters and the following sample size lookup table:⁶²

Power	$\alpha_2 = .01 (\alpha_1 = .005)$										
	d										
	.10	.20	.30	.40	.50	.60	.70	.80	1.00	1.20	1.40
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.70	1924	482	215	122	79	55	41	32	21	15	12
.75	2108	528	236	134	86	60	45	35	23	17	13
.80	2338	586	259	148	95	67	49	38	25	18	14
.85	2611	654	292	165	106	74	55	43	28	20	15
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.99	4808	1203	536	302	194	136	100	77	50	35	26

Resulted in a estimated required sample size between 57 and 74. Since each of our participants had many measurements, our study far exceeded the required sample size to detect the error rate in vital sign measurement between the wearable device and EHR.

In addition to estimating the sample size for validating the wearable data, we also estimated the amount of data necessary for neural network convergence. The choice of cohort size was made based on other clinical studies with similar architectures,^{59, 60} our prior studies,⁶¹ as well as studies on VC dimensions of neural networks.⁵⁸ With this context in mind, we observed that LSTM RNNs typically converge on sample sizes with over 600 patients. Recruiting 888 patients was based on this consideration and the availability of patches during this study.

There does appear to be a study protocol but I could not access this. I am sorry but I found the paper hard to process, possibly partly because the aims are quite broad. One suggestion might be to follow a reporting guideline such as TRIPOD, though this only deals with part of the aims of the study.

- We'd like thank the reviewer for this valuable feedback. To address the reviewer's concerns we have followed the MI-CLAIM checklist (<https://www.nature.com/articles/s41591-020-1041-y>), have included this document in our resubmission and have added the following section entitled "Model Training and Testing" to our Methods section to better describe our study protocol around validating our multivariate prediction model and thank the reviewer for their suggestion (Line 241-272, Paragraph 3):

Model training and testing

We trained a recurrent neural network (RNN) with long-short-term-memory (LSTM) units model that incorporated age, BMI, SBP, and continuously monitored vital signs (HR, RR,

temperature, SpO2, MEWS value, and motion signals) to predict in-hospital deterioration with a 0.5-24 hour time horizon using a 5 hour sequence (Figure 2). We constructed the clinical alert labels detailed previously, based on smoothed MEWS values using a 30-minute-wide rolling average window.

In order to select a model to test on clinical alerts and hard outcomes we performed hyperparameter optimization across the following parameters: number of LSTM layers, sequence length, time horizon, and censor length. The model selected showed superior performance across the various validation steps performed in this study.

Class imbalance was addressed by first randomly removing samples from the negative majority class and oversampling the positive class until we achieve a balance of 95%/5% negative/positive classes of the training set, instead of the original 99.7%/0.3% class balance.

We evaluated the predictive performance of this model in three stages: retrospective, prospective, and alternate Device validations. In the first retrospective validation stage, we validated the model using 5-fold stratified shuffle split cross-validation at the patient level, i.e., no patients in the training set were included in the test set. This resulted in an approximately 80%/20% train/test split. We then compared predictive performance across all folds, using common model validation metrics (ROC AUC, PR AUC) and averaged the resulting ROC and PR curves. After the initial validation stage, we then tested the model in two additional validation stages using two unique datasets: 1) data from prospectively collected data from Device #1 at a separate hospital, and 2) data from a second wearable device used on a different patient population to function as an alternate device test set. In each subsequent stage, the model that was trained on Device #1 data was tested on each dataset. Neither data from prospective Device #1 data nor Alternate Device #2 was used to train the model. These datasets therefore functioned as two true unique test sets.

These three validation stages contrasted performance of the LSTM RNN model against logistic regression (LR) using two concurrently collected datasets: 1) CM data from the wearable device and EHR data collected alongside that wearable data (EHR).

The authors report the agreement between manual and wearable readings – noting that this is poor for respiratory rate. This agreement appears to be used both to decide what data to use for modelling and to assess whether the wearable devices are a viable alternative to manual measurements. They also report Retrospective and Prospective Device #1 and Alternate Device #2 temperature had 99% of all datasets' and devices' measurements fall within the 5% deviation line. This would be surprising for comparisons of skin surface and "Core" (assuming ear probe temperature measurement in the EHR – which is not reported), unless I have misunderstood what a 5% deviation line means. Again, unless I have misunderstood, the Bland-Altman plots seem to show that the 95% limits of agreement are above 1 degree celsius – which would be difficult in clinical practice.

- The author is correct to point out that the skin surface temperature should not fall within 1 degree Celsius of "Core" temperature. Here we'd like to point out that the wearable device provided both a skin temperature measurement and a "body" temperature measurement, which reflects "Core" body temperature. In addition, upon placement of the patch, the body temperature is calibrated by the nurse. For the Bland-Altman plots we used the latter measurement from the device. To make this more clear we've made the following modification to the text (Line 115-119, Paragraph 2):

It should be noted that the wearable device provided both a skin temperature measurement and body measurement, which reflects core body temperature. Upon placement of the patch, the body temperature is calibrated by the nurse. For the wearable data validation we used the body temperature measurement from the device.

I think the authors have then derived a model using a definition of instability of MEWS greater than 6 for ≥ 30 minutes – which I presume is derived from wearable data.

- Thank you for pointing out this important missing detail. Reviewer #1 also pointed out this out. We have added a line to state this clearly in the section “Predictive deterioration modeling; *Model Training and Testing*” (Line 242-247, Paragraph 3):

We trained a recurrent neural network (RNN) with long-short-term-memory (LSTM) units model that incorporated age, BMI, SBP, and continuously monitored vital signs (HR, RR, temperature, SpO₂, MEWS value, and motion signals) to predict in-hospital deterioration with a 0.5-24 hour time horizon using a 5 hour sequence (Figure 2). We constructed the clinical alert labels detailed previously, based on smoothed MEWS values using a 30-minute-wide rolling average window.

They note that the wearable systems detect many more of these episodes than the EPR recordings – and that this is mainly due to differences in wearable recorded respiratory rate. I don't think its possible in this design to determine the usefulness or otherwise of these additional alerts.

- That is correct, it is not possible to determine usefulness of the additional alerts from this data. This dataset is not intended to comment on the clinical utility of the additional wearable generated alerts, since all of the additional alerts were generated post-hoc and offline. We have added this point in the discussion (Line 492-494, Paragraph 5):

It should be noted however, that this study was not intended to comment on the clinical utility of the additional wearable generated alerts, since all of the additional alerts were generated post-hoc and offline.

They then test the model against hard outcomes. I cannot determine whether there were 17 in 17 different patients as suggested by table 1 or 11 as suggested in table 3 – it would seem unlikely that some of these outcomes didn't occur in the same patient – though possibly they may have used the first outcome that occurred in a patient?

Table 1 numbers refer to unique patients who had at least one clinical alert (reached MEWS > 6 for 30 minutes or more). Reviewer is correct that the actual number of these outcomes was higher, since many patients had more more than one clinical alert. The hard outcome analysis – which is a separate validation of the proposed algorithm, included 11 patients (Table 3) that refers to the number of hard outcomes that occurred in the Retrospective Device #1 validation set. As mentioned, these two sets of outcomes were separate from each other. We understand that this might be confusing, so to make this clearer, we've added the following text to the manuscript (Line 277-279, Paragraph 2):

In total 11 hard outcomes were found in the Retrospective Device #1 dataset with sufficient preceding wearable data in order to make predictions. We also ensured that the hard outcomes being predicted upon were at least 24 hours away from each other.

I was not entirely sure how the lack of SpO2 in prospective device 1 was dealt with in prospective evaluation

- Thank you for pointing out this oversight. Reviewer #1 also pointed this out. SpO2 was still included in the final model and the SpO2 variable was imputed with all zeroes. This has been indicated in the section “Predictive deterioration modeling; *Artifact rejection, signal processing and model selection*” (Line 238-239, Paragraph 2):

For the prospective Device #1, SpO2 was not available so the variables was imputed with all zeroes for the final model.

The small number of events (presumably across both devices) is a problem in assessing the usefulness of wearable devices but is problematic for validating a model. Perhaps it would be better to present the work as a feasibility study?

- We agree with both Reviewer #2 and #3 that the small number of clinical hard outcomes is a problem in both assessing utility of wearables and validation of a model. We would like to point Reviewer #2 to Lines 528-529 Paragraph 4 where we explicitly state that this is a “*preliminary capability*.”

“We also showed *preliminary capability* of the model in predicting a broad class of “hard outcomes” against which the model was not trained.”

And in the limitations section in the final paragraph of the discussion where we explicitly acknowledge this is a “*limited validation*” (Lines 561-563, Paragraph 3):

“While this small number made training a model specifically for said outcomes infeasible, our limited validation of the deep learning algorithm on these events showcases that this approach could be applicable to prediction of hard outcomes”

However, to assuage both the reviewers’ concerns we’ve added stronger language to the limitation paragraph to directly take this into account (Line 563-568, Paragraph 3):

It should be noted that, while the limited validation on the hard outcomes is promising, the small number of events just points to feasibility of developing a model capable predicting hard outcomes. However, no strong conclusions can be made. Collecting more data with more patients and more hard clinical outcomes will enable a better validation of this algorithm on hard outcome prediction capabilities.

Reviewer #3 (Remarks to the Author):

The authors have conducted a thorough multi-validation investigation for a wearable based rnn tool for detecting early onset of inpatient deterioration. Overall, a good study but still some major comments that need addressing.

Some comments:

1. Please provide a table with what EHR data and what CM or wearable data provides in terms of variables. easier to refer to and explain

- Thank you for this feedback. We have added the following lines and a table in the methods section “Data sources, patient population, eligibility criteria and sample size calculation” that makes the data sources for each of the variables clearer (Line 123-128):

Table 1a summarized data sources for each of the continuous monitoring and demographics variables

■ CM ■ EHR	Retrospective Device #1	Alternate Device #2	Prospective Device #1
HR			
RR			
Temp			
SpO2			N/A
SBP			
Age			
BMI			
SMA		N/A	
MEWS			

Abbreviations: CM = Continuous Monitoring device; EHR = Electronic Health Record; HR = heart rate; RR = respiratory rate; Temp = temperature; SpO2 = oxygen saturation; SBP = systolic blood pressure; BMI = body mass index; SMA = signal magnitude area; Modified Early Warning Score (MEWS).

2. mews score is not clear to me. please describe in clearer mathematical language. how does the metric work - intuition is not clear either.

- Thank you for pointing out this oversight. We have added the following additional wording and equation to more clearly explain the MEWS score (Line 190-198, Paragraph 3):

The MEWS calculation is based upon vital sign measurements given by the equation below:

$$MEWS = SBP^{Score} + HR^{Score} + RR^{Score} + Temp^{Score} + AVPU^{Score} + Age^{Score} + BMI^{Score}$$

Depending on the range of each item, it is given a score ranging from 0 to 3 (Supplementary Table 2), and the final score is tallied (range 0-15), with higher values indicating greater risk of clinical decompensation. It should be noted that this a commonly used score, not only in our system, but in health systems around the world, as an early warning system.

3. a figure on a hypothetical patient for prediction would be useful. i.e. here is patient values, at

some point is “baseline” or start of follow-up and some point is the event deemed so by episodic monitoring (i believe?)

- We have added the following Supplementary Figure 4 that gives an example of the patients values that preceded a ICU transfer event (Lines 804-815):



Supplementary Figure 4. Sample patient vitals preceding an ICU transfer event.

Sample patient vitals preceding an ICU transfer event are shown of: manually measured vital values input by a nurse into the EHR (red traces), the most recently calculated MEWS score based on all of the manually measured vitals in the EHR (red trace, top plot), the most recently measured wearable value for that particular vital (blue traces), and the most recently calculated MEWS score based on the continuously measured vital signals from the wearable device (blue traces, top plot). The black dotted line in the top plot represents the MEWS 6 threshold that was used for identifying clinical alerts. The vertical magenta line indicates the time of the ICU transfer event. The green shaded area represents an example 5-hour sequence of data that is extracted and passed into a model to try to predict the ICU transfer event, in this case with a three-hour lead time.

4. clearly describe what is the outcome being predicted in mews score section (ie., > 30-minute periods of elevated MEWS, ...)

- Thank you for pointing out this important missing detail. Both reviewers #1 and #2 pointed this out. We have added a paragraph to state this clearly in the section “Predictive deterioration modeling; *Model training and testing*” (Line 242-247, Paragraph 3):

We trained a recurrent neural network (RNN) with long-short-term-memory (LSTM) units model that incorporated age, BMI, SBP, and continuously monitored vital signs (HR, RR, temperature, SpO2, MEWS value, and motion signals) to predict in-hospital deterioration

with a 0.5-24 hour time horizon using a 5 hour sequence (Figure 2). We constructed the clinical alert labels detailed previously, based on smoothed MEWS values using a 30-minute-wide rolling average window.”

5. I am a fan of the wearable data evaluation. as RR was inaccurately measured, could there be some scaling/calibrating that can make the the measurements fall within the 10% deviation line? that way you use as much data as possible...

- Thank you for the kind words. We did in fact use the Respiratory Rate from the wearable device in the modelling because we had a good reason to believe that it was closer to the ground truth than the Respiratory Rate in the EHR based on other wearable validation studies.⁴⁶

In addition, we'd like to point the reviewer to the following points in the discussion on respiratory rate (Lines 486-490, Paragraph 5):

“We also investigated the driving components that led to these additional wearable-based alerts and found that respiratory rate and heart rate were the two major factors. This is not a surprising result for RR, since inaccurate documentation of RR in the EHR is well known due to digit preference and the increased burden of measuring and reporting an accurate RR.⁴⁹”

6. please put the details on logistic regression and other benchmark model in the method section. what variables etc?

- Thank you for pointing out this important detail. Reviewer #2 raised this concern as well. We have added the following paragraph to the section “Predictive Deterioration modelling; Model training and testing” to fill-in these important details (Line 242-247, Paragraph 3):

We trained recurrent neural network (RNN) and logistic regression models that incorporated age, BMI, SBP, and continuously monitored vital signs (HR, RR, temperature, SpO2, MEWS value, and motion signals) to predict in-hospital deterioration with a 0.5-24 hour time horizon using a 5 hour sequence (Figure 2). We constructed the clinical alert labels, detailed previously, based on smoothed MEWS values using a 30-minute-wide rolling average window.

7. in alternative device, why do you think auroc is higher for log reg - ehr model?

8. this is quite odd. please include discussion material on this... and potentially text in limitation section as well.

9. also, device 1 prospective validation shows that ehr model LR has comparable AUC w respect to rnn model with wearables data - 0.89 v 0.9. only Device 1 retrospective validation shows promise for the proposed methods. given this, generalisability of method is not certain. please discuss.

- We agree with the reviewer that by itself the higher AUROCs for the Device #2 EHR-LR model and the Device #1 Prospective validation EHR-LR is rather odd. These anomalies are a good example of why the inclusion of the Precision-Recall curve is important—the ROC plot can be misleading on the performance of a classifier on a highly imbalanced dataset such as the one in this study (99.7% / 0.3% negative /

positive class balance). For more details on how we address this peculiar behavior, please see following paragraph (Line 506-516, Paragraph 2):

We'd also like to note the higher, and comparable, AUROCs for the Device #2, and Device #1, logistic regression EHR models, respectively. These anomalies are a good example of why the inclusion of the Precision-Recall curve is important—the ROC plot can be misleading on the performance of a classifier on a highly imbalanced dataset such as the one in this study (99.7% / 0.3% negative / positive class balance). Given this, we'd like to point out that the RNN significantly outperformed the logistic regression model on the precision-recall curve for the Prospective Device # 1 validation and both EHR and wearable data being comparable for the Device # 2 validation. This discrepancy captures the fact that the logistic regression model simply performed better than the RNN on predicting true negatives (hence the higher AU-ROC), however the RNN was still significantly superior on predicting true positives (hence the higher AU-PR).

10. not enough events in table 3 to make any call here... huge limitation that needs to be taken into account - would be better if removed.

- We agree with the reviewer that the limited number of events is a significant limitation in this table. Reviewer #2 also flagged this as an important limitation. We would like to point the reviewer to Line 528-529 Paragraph 4 where we explicitly state that this is a "*preliminary capability*:"

"We also showed *preliminary capability* of the model in predicting a broad class of "hard outcomes" against which the model was not trained."

And in the limitations section in the final paragraph of the discussion where we explicitly acknowledge this is a "*limited validation*" (Line 561-563, Paragraph 3):

"While this small number made training a model specifically for said outcomes infeasible, our limited validation of the deep learning algorithm on these events showcases that this approach could be applicable to prediction of hard outcomes"

However, to assuage both reviewers' #2 and #3 concerns we've added stronger language to the limitation paragraph to directly take this into account (Line 563-568, Paragraph 3):

It should be noted that, while the limited validation on the hard outcomes is promising, the small number of events just points to feasibility of developing a model capable predicting hard outcomes. However, no strong conclusions can be made. Collecting more data with more patients and more hard clinical outcomes will enable a better validation of this algorithm on hard outcome prediction capabilities.

REVIEWER COMMENTS

We wish to thank the reviewers again for this second round of thoughtful comments and suggestions. As you will see in this document, as well as the revised manuscript, the specific points raised by the reviewers have all been addressed; clarifications and edits to the abstract, methods, results and discussion sections, as well as a table have been added to supplementary information. We believe all this additional work has significantly improved the paper and we hope the revisions are satisfactory to the reviewers and the editors. Our responses are highlighted in blue and the corresponding additions/changes to the revised manuscript are highlighted in red. All line numbers refer to the red-lined manuscript.

Reviewer #1 (Remarks to the Author):

Authors have made a significant effort to address comments from all reviewers. I have no further concerns for this manuscript to be published.

- Thank you for the kind words and for the invaluable feedback.

Reviewer #2 (Remarks to the Author):

The authors have made good progress with answering the comments of the reviewers.

I would suggest that the abstract is clarified to ensure the reader understands that patients were included as part of a quality initiative, and prediction models have been retrospectively built and validated on these data. It should also be clear that the authors experimented with different data periods for prediction to create best model performance.

- Thank you for suggesting this clarification. We have made the following edits to the abstract to make the above point about patients being part of a quality initiative (Lines 27 – 34) :

Using data from 888 adult non-ICU inpatient visits in four hospitals in New York and employing two different clinical grade wearable biosensors (4-8% data missingness, excluding SpO2), as part of a quality initiative, we trained a recurrent neural network (RNN) to predict both MEWS alerts and adverse clinical outcomes. Using multiple stages of validation, we showed in our retrospective, time-sequence duration optimized, prospective validation the RNN model was able to predict both periods of elevated MEWS scores (ROC AUC 0.89 +/- 0.3, PR AUC 0.58 +/- 0.14) and adverse clinical outcomes (accuracy: 81.8% on 11 events) up to an average of 17 hours in advance.

I think the use of "prospective" could unintentionally mislead the reader in the context of work done on a retrospective dataset. I think a retrospective dataset has been split to allow model development with unseen data held out for testing. If this is the case I think it would be better if this was made clear throughout.

- We would like to point the reviewer to the following clarifications (emphasis on bolded section) that were added in the first round of reviews on Lines 265 – 275:

After the initial validation stage, we then tested the model in two additional validation stages using two unique datasets: **1) data from prospectively collected data from Device #1 at a separate hospital**, and 2) data from a second wearable device used on a different patient population to function as an alternate device test set. **In each subsequent stage, the model that was trained on Device #1 data was tested on each dataset. Neither data from prospective Device #1 data nor Alternate Device #2 was used to train the model. These datasets therefore functioned as two true unique test sets.**

We would like to emphasize that the “prospective” dataset was a true prospective dataset in the sense that it was collected 1) chronologically entirely after the retrospective dataset, that the model was trained on 2) from an entirely separate cohort of patients at a different hospital than the training dataset and 3) no data from the prospective dataset was used to optimize or test the model that was built on the retrospective data.

To make this clearer, the following edits have been made to Lines 265 – 275:

After the initial validation stage, we then tested the model in two additional validation stages using two unique datasets: 1) data collected **prospectively** from Device #1, **i.e. entirely after Device #1's retrospective data used to train the model**, at a separate hospital **and from a different patient cohort**, and 2) data from a second wearable device **used at a separate hospital and on an entirely** different patient population to function as an alternate device test set. In each subsequent stage of validation, **i.e. “Prospective” and “Alternate Device” validations**, the model **that was trained exclusively on Device #1's retrospective dataset** was **then** tested on each of these **pure validation** datasets. Neither data from Prospective Device #1 data nor Alternate Device #2 was used to **optimize, hyperparameter tune or train** the **retrospectively trained** model **in anyway**. These datasets, **and their accompanying validation results**, therefore functioned as two true unique test sets **for the retrospectively optimized and trained model**.

Given the small number of events - prediction of events is at best initial work as the other reviewers note. If they are to be included in the abstract the small number of events should be made clear (81% of ? 11 events).

- We'd like to thank Reviewer #2 for this observation and agree that this is indeed initial work. To address these concerns, we've made the following change to the abstract (Line XX – XX):

Using multiple stages of validation, we showed the RNN model was able to predict both periods of elevated MEWS scores (ROC AUC 0.89 +/- 0.3, PR AUC 0.58 +/- 0.14) and adverse clinical outcomes (accuracy: 81.8% of 11 events) up to an average of 17 hours in advance.

The abstract should also include the proportion of time when monitors were applied where sufficient data quality was available for model training/application.

- We agree with the reviewer on this point and have added Supplementary table 5, detailing the missingness for each vital signal in the data.

Missingness (%)	VitalPatch	Biobeat
Respiratory Rate	3.72	8.3
Heart Rate	3.72	0
Body Temp.	0	3.9
SpO2	14.95	23.02
Activity (SMA)	0	0

Supplementary Table 5. Missing data from CM devices from preliminary data from 888 patients

And have made the following edit to the abstract (Lines 27 - 30):

Using data from 888 adult non-ICU inpatient visits in four hospitals in New York and employing two different clinical grade wearable biosensors (4-8% data missingness, excluding SpO2), as part of a quality initiative, we trained a recurrent neural network (RNN) to predict both MEWS alerts and adverse clinical outcomes.

I remain unclear how temperature agreements with 95% confidence intervals above 1 degree would be useful in clinical practice and I think unless I have misunderstood this this limitation needs clarifying. Presumably as the patch was calibrated by the nurse the first ? number of readings were excluded as they would match as a result of calibration?

- We agree with the reviewer that by itself, a temperature reading with a 95% confidence interval above 1 degree would not be useful in clinical practice. However, the goal in this paper was to capture trends via continuous monitoring wearable devices to predict adverse clinical events. It is not our intention in this paper to try to justify the use of raw temperature readings from clinical wearables in clinical decision making. Since the vitals that went into the predictive algorithm were normalized, the absolute error was not as important as the trends. To clarify this point, we've made the following edits on lines 497 – 503:

It should be noted that, while certain vital signals had an error magnitude that might preclude its use in clinical decision making (e.g. temperature), the goal in this paper was to capture trends via continuous monitoring wearable devices to predict adverse clinical events. It was not our intention in this paper to try to justify the use of raw vital signal readings from clinical wearables for use in clinical decision making. Since the vitals that went into the predictive algorithm were normalized, the absolute error of each vital signal was therefore not as important as the trends.

However, with these suggested clarifications the work is interesting and has been improved by the thoughtful responses to the contributions of the other reviewers. It's a lot of work and the authors are to be congratulated on a thoughtful approach to a taxing problem

- We'd like to thank the reviewer again for their thoughtful and insightful comments. We believe these comments have definitively improved the rigor of the work presented in this paper and hope they are to the satisfaction of the reviewer.

Reviewer #3 (Remarks to the Author):

The authors of the "Beyond episodic early warning systems: a continuous clinical alert system for early detection of in-hospital deterioration" have done a remarkable job editing and revising the manuscript. Very thorough work.

- Thank you for this high praise and for the invaluable feedback that has helped improve the quality of this work.

They have responded to my reviewer 3 comments pretty comprehensively. I appreciate all the work they have done adding figures/tables and text that help explain the prediction objective, baseline, cohort, etc. However I still have some concerns on the utility and impact of the proposed model.

Comments:

1. "The patients that were patched were selected based often on the best clinical judgement of the staff in each of the wards where the wearable devices were deployed. In practice, this meant the nurse overseeing the patient would often decide whether to have a wearable device placed on the patient or not based on their assessment of how at risk the patient was for

decompensating.” This means that the selection of patients is subject to selection biases. What were the conditions they were chosen? In a really troubling hypothetical setting, patients were chosen in such a way that would “benefit” the model?

- We’d like to thank Reviewer #3 for this important observation. We have added the following paragraph to the discussion to expatiate on this point (Lines 481 – 490):

To address the concern of a selection bias existing in our patient selection process, we’d like to point out our deterioration event rate of ~1% was actually lower than what is seen in the clinical literature (~6%)¹. While clinicians were basing the patch decision on their best clinical judgement, this didn’t lead to an increased deterioration event rate, for a variety of reasons, either due to: the acuity of the wards, the patching occurring outside of acute time periods, or nurses patching patients based on assessment of risk while also waiting until a convenient time to patch them. If a selection bias existed that benefitted the model, i.e. more deterioration event labels and more balanced data for the model to train more effectively on, we would expect to see higher rates of deterioration, which we don’t see. The patching scheme thus reflected real world conditions and therefore a real world scenario of the patching situation.

2. Also how generalizable is the model to the general population of patients for which you want to put the patch on and the wearables device? Yes indeed the model has been validated not on a different cohort but an alternate device. This is not the same as cohort being different.

- We’d like to thank Reviewer # 3 for pointing out the lack of clarity in our description of our validation sets, patient cohorts for each validation set and subsequent validation steps. The cohort of patients that made up Alternate Device #2 data, was indeed entirely different from the cohort of patients that made up Retrospective Device #1’s data, on which the model was trained. Reviewer #2 had similar concerns and we’ve made the following edits to clarify our definition of the validation sets (relevant edits to Reviewer #3’s above comment bolded below, Lines 265 – 275):

After the initial validation stage, we then tested the model in two additional validation stages using two unique datasets: 1) data collected **prospectively** from Device #1, i.e. **entirely after Device #1’s retrospective data used to train the model**, at a separate hospital **and from a different patient cohort**, and **2) data from a second wearable device used at a separate hospital and on an entirely different patient population to function as an alternate device test set**. In each subsequent stage of validation, i.e. “Prospective Device #1” and “Alternate Device #2” validations, the model **that was trained exclusively on Device #1’s retrospective dataset** was **then** tested on each of these **pure validation** datasets. Neither data from prospective Device #1 data nor Alternate Device #2 was used to **optimize, hyperparameter tune or train the retrospectively trained model in anyway**. These datasets, **and their accompanying validation results**, therefore functioned as two true unique test sets **for the retrospectively optimized and trained model**.

3. “Class imbalance was addressed by first randomly removing samples from the negative majority class and oversampling the positive class until we achieve a balance of 95%/5% negative/positive classes of the training set, instead of the original 99.7%/0.3% class balance. “ this could be really dangerous for calibration. Please check: <https://arxiv.org/html/2404.19494v1>

- We'd like to thank Reviewer #3 for raising this important point regarding rebalancing. We agree that in the context of rebalancing for more traditional ML models, such as the decision trees, logistic regression and SVM models, as in the linked reference, rebalancing can indeed create scenarios of poorly calibrated models. However, we have not seen this same effect of addressing class imbalance on the calibration of LSTM RNNs in the literature and instead have seen the LSTM RNNs remain well calibrated after rebalancing.^{1,2} Also, while calibration is an important performance metric that needs to be monitored on models that provide probability estimates (risk prediction/estimation), and some of our prior work has focused on exactly the importance of monitoring calibration^{3,4}, this specific work is focused on providing a binary output, a flag for deterioration and not a risk estimate, thus calibration does not apply as a performance metric (more on our response to the following point).

4. Given this, I recommend tests of calibration. Please conduct.

- Thank you for this suggestion. As mentioned, since we do not provide a risk estimate, but rather a binary flag for predicted deterioration, since we have not framed this problem in order to provide a sliding probability scale of deterioration, i.e. a “risk score,” using the LSTM RNN, so a calibration plot would not be as informative as the ROC and Precision-Recall curves that we've provided in our results. Moreover, including the calibration curve would confuse the reader that we're providing a deterioration probability or “risk score” when we're not. Lastly, in our model, probability of deterioration is not estimated at any point in our architecture (even at the final layers that do not have one final cell that gets thresholded), thus we cannot create a risk estimation tool and a calibration curve for that.

5. I like the paragraph on how the RNN v LR is similar for AUROC but not for AUPRC. Great.

- Thank you for the compliment, we appreciated the feedback about the specific behavior of the ROC curve performances that prompted this edit, we have seen similar behavior of ROC curves in other studies where the AUROC is similar but PR and calibration curves are not³.

Reviewer #3 (Remarks on code availability):

I believe no code available.

- We've made code available for reviewers at this link:

- Code Ocean link:
 - <https://codeocean.com/capsule/9888403/tree>

1. Acharya, T., Annamalai, A., & Chouikha, M. F. (2023). Optimizing the Performance of Network Anomaly Detection Using Bidirectional Long Short-Term Memory (Bi-LSTM) and Over-sampling for Imbalance Network Traffic Data. *Advances in Science, Technology and Engineering Systems Journal*, 8(6), 144-154.
2. Zwerwer, L. R., Luz, C. F., Soudis, D., Giudice, N., Nijsten, M. W., Glasner, C., ... & Sinha, B. (2024). Identifying the need for infection-related consultations in intensive care patients using machine learning models. *Scientific Reports*, 14(1), 2317.
3. Levy, T. J., Coppa, K., Cang, J., Barnaby, D. P., Paradis, M. D., Cohen S. L., ... & Zanos, T. P. (2022). Development and validation of self-monitoring auto-updating prognostic models of survival for hospitalized COVID-19 patients. *Nature Communications* 13, 6812.
4. van Klaveren, D., Zanos, T. P., Nelson, J., Levy, T. J., Park, J. G., ... & Kent, D.M. (2022). Prognostic models for COVID-19 needed updating to warrant transportability over time and space. *BMC Med* 20, 456.

REVIEWER COMMENTS

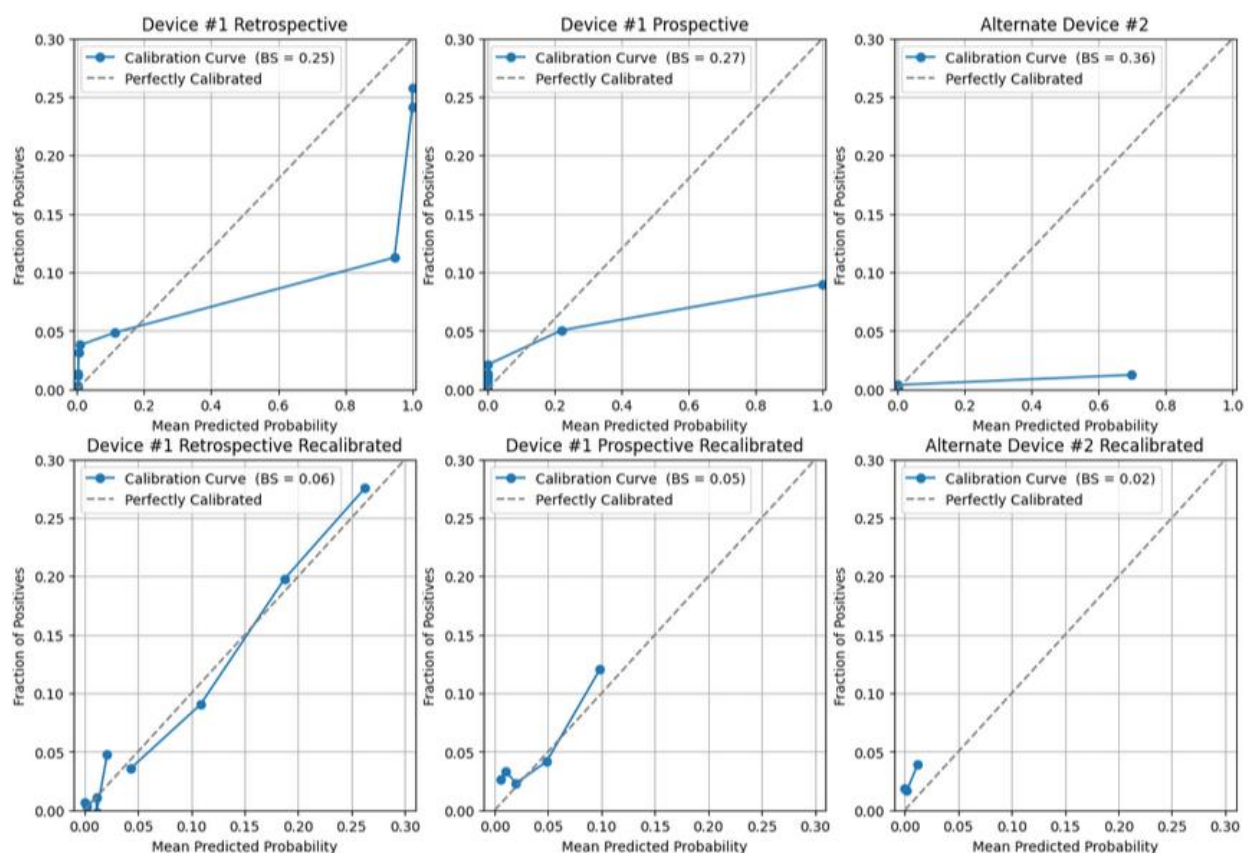
We wish to thank the reviewers again for this second round of thoughtful comments and suggestions. As you will see in this document, as well as the revised manuscript, the specific points raised by the reviewer have all been addressed; clarifications and additional text has been added in the methods and discussion sections, and an additional figure has been added to the supplementary information. We believe all this additional work has significantly improved the paper and we hope the revisions are satisfactory to the reviewer and the editors. Our responses are highlighted in blue and the corresponding additions/changes to the revised manuscript are highlighted in red. All line numbers refer to the red-lined manuscript.

Reviewer #3 (Remarks to the Author):

The answers to most of my previous comments were fine. However, the responses to the ones on calibration were far from acceptable in my eyes. Calibration is key for showing reliability of modelling.

1. Indeed you have a binary output, but this binary output is derived from continuous estimates. Otherwise you could not present confusion matrices and area under PRC - the latter for sure can be estimated as precision-recall estimates across the spectrum of risk threshold. So if you can't get risk estimates, then I ask how did you derive area under PRC? In other words, it is clear there is a continuous risk prediction happening here and there is binary output calculation following continuous risk estimation based on threshold - e.g., >0.8 is 1 and ≤ 0.8 is 0

We thank the reviewer for their persistence on this important point, which prompted additional analysis and some important insights into calibration performance of not only our model, but in general neural network architectures such as LSTM RNN. Reviewer is correct that there is a continuous output given by the deep neural network. In order to address the comment, we've calculated calibration curves and Brier scores and included the following figure of original and recalibrated calibration curves with Brier Scores as Supplementary Figure 5 (Lines 875 – 881):



Supplementary Figure 5. Calibration Curves for the LSTM RNN Model

Calibration curves (blue traces) across the three stages of validation Device #1 Retrospective (left), Device #1 Prospective (center) and Alternate Device #2 (right). The first row shows original calibration curves for each stage of validation from left to right. The second row shows the recalibrated calibration curves for each stage of validation from left to right. Brier scores (BS) are included in the figure caption. Perfect calibration (dotted line) is shown in each case.

The LSTM RNN model calibration results above, while not optimal, are not surprising. It is known that probabilistic predictions, are particularly challenging for Neural Network architectures, including LSTM RNN models, that are known to exhibit overconfidence, pushing predicted probabilities toward 0 or 1, mostly arising from the interplay between activation functions and unregularized logit magnitudes. The bi-modal distribution of outputs that amplifies the distance between points at the decision boundary between lower and higher risk patients is inherent to how LSTM model achieve superior discriminative power, however it does have an impact on calibration. In our Retrospective Device #1 calibration curve this is exactly what we see (“Device #1 Retrospective” Supplementary Figure. 5). Between approximately a 5% and 10% fraction of positives, the mean predicted probability jumps from 10% to >95%, this means the model has set it’s decision boundary at a point in the input space corresponding to

approximately a 7% true risk of deterioration (i.e. fraction of positives) and scales the risk exponentially as one crosses this boundary.

Furthermore, given the multiple nested non-linear activation functions inherent in a deep neural network LSTM architecture, it is not surprising that these model types produce a non-continuous, non-linear numerical output that does not correspond to an estimate of the risk of deterioration (see Kumar et al. ICML 2018). While our choice for not providing probability risk estimates as an output of the proposed model was not driven by this insight, it is now even more pertinent that we avoid providing (even recalibrated) probability estimated risks of deterioration but rather a binary flag (our motivation to stick to a flag is that it provides a more clear and actionable path to mitigating strategies, such as an ICU consult and elevation of care). While it is evident from the calibration curve and Brier scores that the model is not well calibrated initially, the main goal of the model, to exhibit strong discriminatory ability, is still maintained, as evident by the ROC and PR of the model. At the same time, these new results establish that our approach is not optimal to be used as a risk stratification model, which was not intended in the first place.

As mentioned, we do believe that this additional analysis and the results are important, so we have also added the following section to the discussion, addressing the results of the calibration curve (Lines 556 – 576) and some of the insights gained:

To assess model calibration, we looked at calibration curves of the LSTM RNN model for each stage of validation. The results are not surprising, given that modern neural network architectures are known to exhibit overconfidence, pushing predicted probabilities toward 0 or 1, mostly arising from the interplay between softmax/sigmoid activations and unregularized logit magnitudes. This “confidence sharpening” is evident in our Retrospective Device #1 calibration curve (“Device #1 Retrospective” Supplementary Figure. 5). Between approximately a 5% and 10% fraction of positives, the mean predicted probability jumps from 10% to >95%, since the model has set its decision boundary at a point in the input space corresponding to approximately a 7% true risk of deterioration (i.e. fraction of positives) and scales the risk exponentially as one crosses this boundary. The same behavior is seen in the prospective validation calibration curve, however since the dataset is smaller and there are less acute patients, the highest risk patients in this dataset are only at a ~10% risk of deterioration as opposed to the ~25% seen in the retrospective dataset. The Alternate Device #2 dataset takes this point even further in that the highest risk group of patients only have ~1-2% chance of deterioration. Additionally, given the multiple nested non-linear activation functions inherent in a deep neural network LSTM architecture, it is not surprising that this produces a non-continuous, non-linear numerical output that does not correspond to an estimate of the risk of deterioration. While it is evident from the calibration curve and Brier scores that the model is not well initially well calibrated, this does not undermine our findings in the ROC and PR of the model’s strong discriminatory ability. At the same time, this establishes that this is not a risk stratification model, but rather a binary classification model that produces an actionable output, flagging patients for ICU consults and possible elevation of care. Finally, recalibration can correct model calibration across the observed ranges of patient risk.

Further we’ve added the following section to the Methods (Line 281 – 284) :

Model Calibration

Calibration curves were constructed for the LSTM RNN model across the three stages of validation (Supplementary Figure. 5). Isotonic regression recalibration was performed on the calibration curves for each stage of validation. Brier scores (BS) were also calculated on all calibration curves.

We hope that the above responses and edits are to the satisfaction of Reviewer #3 regarding the calibration curves. We'd also like to point the reviewer to a pair of similar studies that showed comparable Brier scores in the calibration of their LSTM RNN models:

Vagliano, I., Lvova, O., & Schut, M. C. (2021). Interpretable and continuous prediction of acute kidney injury in the intensive care. In *Public Health and Informatics* (pp. 103-107). IOS Press.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8162230/> (table 1 - BS = ~0.1 - 0.2)

Deng, Y., Liu, S., Wang, Z., Wang, Y., Jiang, Y., & Liu, B. (2022). Explainable time-series deep learning models for the prediction of mortality, prolonged length of stay and 30-day readmission in intensive care patients. *Frontiers in Medicine*, 9, 933037.

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9554013/> (table 2 - BS = ~ 0.8 - .185)

2. While resampling may work on select cases without affecting calibration as you have shown in the references, the best way to show this point indeed is to show the calibration. So again, I urge you please show calibration curves.

Reviewer #3 is correct in pointing out that resampling could worsen calibration. We'd like to thank them for their persistence on looking at and addressing this directly. We can see from the calibration curves above that the models are not well calibrated and believe the edits made as a result have strengthened the manuscript.

The clinically relevant goal of this manuscript, established at the outset, was to predict absence or presence of imminent deterioration risk. The study by Welvaars et al. (below) suggests that the resampling we did improves discriminatory performance of the model but worsenes calibration. Since it was not our goal to create a model that outputs a continuous estimate of risk, this is an acceptable tradeoff. Further, we argue that despite the model being poorly calibrated, there is still clinical utility due to the model's robust discriminatory ability of low vs high risk patients for deterioration. Lastly, we'd like to point the Welvaars et al. paper on this exact topic of the effect of resampling on performance and calibration of models, that makes similar claims and arguments to our own:

Welvaars, K., Oosterhoff, J. H., van den Bekerom, M. P., Doornberg, J. N., van Haarst, E. P., & OLVG Urology Consortium, and the Machine Learning Consortium van der Zee JA van Andel GA Lagerveld BW Hovius MC Kauer PC Boevé LMS van der Kuit A Mallee W Poolman R. (2023). Implications of resampling data to address the class imbalance problem (IRCIP): an

evaluation of impact on performance between classification algorithms in medical data. *JAMIA open*, 6(2), ooad033.
<https://academic.oup.com/jamiaopen/article/6/2/ooad033/7187469>

REVIEWER COMMENTS

We wish to thank the reviewer again for this fourth round of thoughtful comments and suggestions. As you will see in this document, as well as the revised manuscript, the specific points raised by the reviewer have all been addressed; clarifications, table edits and figure updates have been added. We believe all this additional work has significantly improved the paper and we hope the revisions are satisfactory to the reviewer and the editors. Our responses are highlighted in blue and the corresponding additions/changes to the revised manuscript are highlighted in red. All line numbers refer to the red-lined manuscript.

Reviewer #3's comments:

Excellent work doing full calibration analyses. Indeed, these are not optimal calibration results, but the transparency in reporting this in results and extended discussion sections is duly appreciated.

- Thank you for the kind words. We have addressed all the helpful comments raised by Reviewer #3.

Some more points:

1. I don't agree with this statement: " At the same time, this establishes that this is not a risk stratification model, but rather a binary classification model that produces an actionable output, flagging patients for ICU consults and possible elevation of care. Finally, recalibration can correct model calibration across the observed ranges of patient risk." When you do binary classification indeed this stratification of risk. Those below the threshold are low risk and those above are high-risk. Hence not sure this is accurate to put in. Please remove.

- We have removed this sentence from the discussion section (Lines 577-579).

2. Great to see recalibration works well and the model looks better calibrated throughout 3 analyses. Has table 2 been changed accordingly? recalibration would have effected the true/neg capture. Not necessary to change but good to note that if table 2 is based off pre- recalibrated estimates, that this is written in footnotes of table next to abbrev?

- We have updated Table 2 with the predictions from the recalibrated model. Please see below:

A) Device #1 Retrospective Thresh = 0.0303				Predicted Label
		No Alert - 0	Alert - 1	
True Label	No Alert - 0	14308	11889	Specificity : 0.55
	Alert - 1	213	3317	Sensitivity : 0.94
		NPV : 0.99	PPV : 0.22	
B) Device #1 Prospective Thresh = 0.0352				Predicted Label
		No Alert - 0	Alert - 1	
True Label	No Alert - 0	2653	2128	Specificity : 0.55
	Alert - 1	14	480	Sensitivity : 0.97
		NPV : 0.99	PPV : 0.18	
C) Alternate Device #2 Thresh = 0.0291				Predicted Label
		No Alert - 0	Alert - 1	
True Label	No Alert - 0	33957	4633	Specificity : 0.88
	Alert - 1	280	597	Sensitivity : 0.68
		NPV : 0.99	PPV : 0.11	

Also for decive 1 prospective and alternate device #2 does this mean the threshold of interest is reset? there are no predictions

above 0.1 for alternate device #2 and hence the threshold of 0.9 would not make much sense now?

- That is exactly right—the thresholds have now been recalibrated to around a 3-4% risk of deterioration for all stages of validation (Retrospective, Prospective and Alternate Device #2). Table 2 has been updated accordingly and we thank Reviewer #3 for this astute clarification. With these results, the recalibration step becomes quite beneficial to the model and we thank the reviewer for improving our model with their suggestions.

3. Also the precision-recall curve would have changed as well. So any notes or thoughts on this?

- Yes, a couple thoughts. Isotonic recalibration can change the AUC of the PR curve in binary classification due to several factors related to the calibration process and its impact on probability rankings. Since PR curves are more sensitive to calibration errors than ROC curves—because precision depends directly on predicted probabilities (not just rankings)—recalibration can improve PR AUCs. Further, since isotonic recalibration adjusts probability magnitudes, it can also improve poorly calibrated probabilities, thus increasing PR-AUC.

With recalibration, I would say all discrimination metrics and analyses that rely on decision thresholds must be redone. While AUROC will remain the same generally, all others would move around.

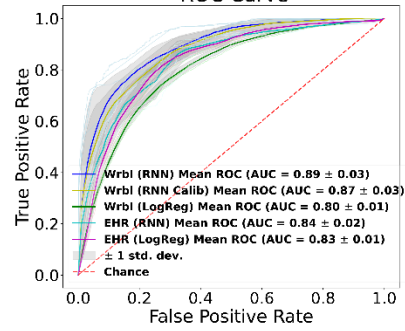
- In our experience, isotonic recalibration can in fact change the AUC of the ROC in binary classification. One would think that isotonic regression, since it is a monotonic transformation, should in theory preserve the relative order of predicted probabilities, meaning the ROC-AUC should remain unchanged. However, isotonic regression can introduce ties in probabilities (e.g., grouping similar predictions into bins), which may slightly alter the ROC curve's shape and AUC. Some examples include cases where many tied probabilities occur near

the decision threshold and hence the ROC curve's granularity decreases. Ties can lead to "flat" regions in the curve, as seen in the recalibrated ROC curve of the Prospective validation, affecting AUC calculations, albeit slightly, as shown in our results.

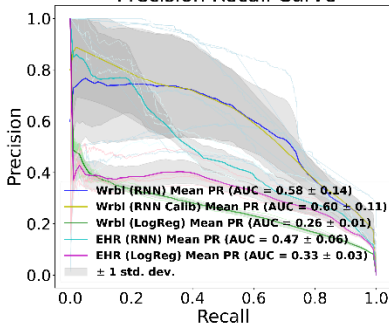
Recalibration is an important step and it is great to see it done. However, we can't see the full picture without all other analyses conducted given recalibrated estimates.

- To address this point we have redone all of the discrimination metrics and analyses in Figure 3, panels A-F (second curve labelled "Wrbl (RNN Calib)..." to show the recalibrated ROC and PR curves. As you might expect, the AUCs of both curves change very slightly, by 1-2% in some cases, or do not change at all (Lines 784-812).

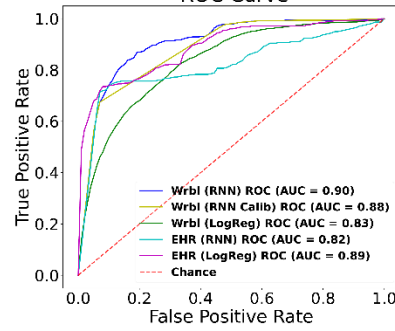
A Device #1 Retrospective Validation ROC Curve



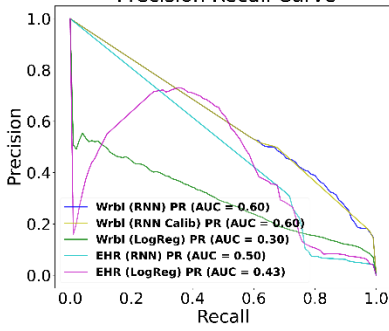
B Device #1 Retrospective Validation Precision Recall Curve



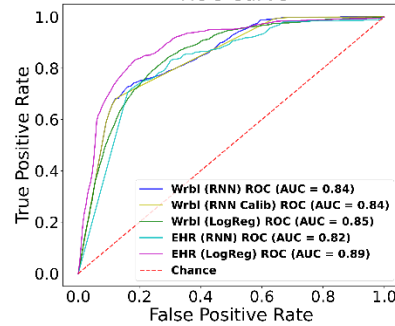
C Device #1 Prospective Validation ROC Curve



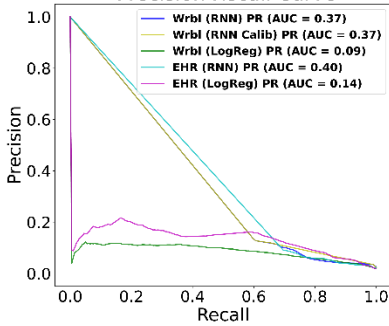
D Device #1 Prospective Validation Precision Recall Curve



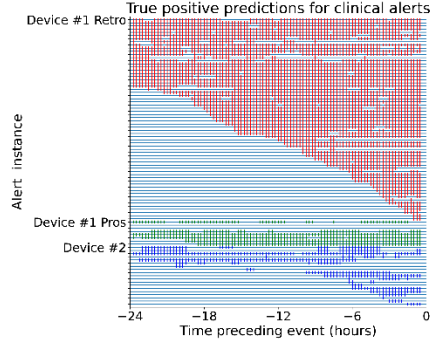
E Device #2 Retrospective Testing ROC Curve



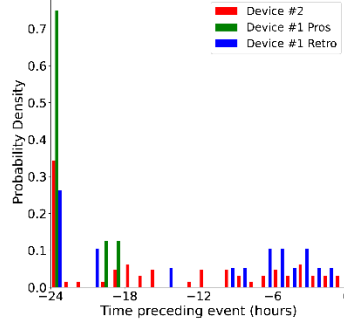
F Device #2 Retrospective Testing Precision Recall Curve



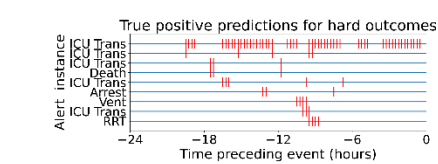
G True positive predictions for clinical alerts



H Lead time distribution for clinical alerts



I True positive predictions for hard outcomes



J Lead time distribution for hard outcomes

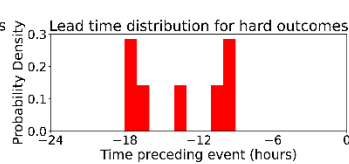


Figure 3. (Panels A-J) Wearable (Wrbl) based Recurrent Neural Network (RNN), **Recalibrated Wearable (Wrbl) based Recurrent Neural Network (RNN Calib)**, Logistic Regression (LogReg) and EHR based model performances illustrated by receiver operator characteristic (ROC) and precision-recall (PR) curves for twenty-four types of validation. (Panels G-J) Lead time analysis of wearable based RNN model illustrates how far in advance model can predict both clinical alerts and hard outcomes.

ROC and PR curves for predicting periods 30mins or greater of continuously elevated MEWS (MEWS > 6) 0.5-24 hours into the future. AUCs for each curve are listed in the figure legend. (Panel A, B) Retrospective RNN, **Recalibrated RNN** and LR model performances in predicting an elevated MEWS using randomly shuffled Device #1 data, and concurrently collected EHR data, which was held out for testing. Performance of each model, for each dataset, across folds is plotted (lighter shade). Mean performances are plotted (darker shade) with one standard deviation of performances across folds also plotted (grey). Chance performance of a binary classifier is plotted (dotted red line). (Panel C, D) RNN, **Recalibrated RNN** and LR Model performances in predicting an elevated MEWS using prospective data from the same device (Device #1 Prospective) and concurrently collected EHR data, which were both held out entirely for testing. (Panel E, F) RNN, **Recalibrated RNN** and LR Model performances in predicting an elevated MEWS using data from another device (Alternate Device #2), and concurrently collected EHR data, which were both held out entirely for testing. Lead times here are the timing of the first true positive preceding an alert. 'ICU Trans' = Unplanned ICU Transfer, 'Arrest' = Cardiac Arrest, 'Vent' = Ventilator, 'RRT' = Rapid response team call (Panel G) Raster plot visualization of timing of all true positive predictions made by RNN model (**not recalibrated**) with wearable data preceding a clinical alert (Time 0). The colors represent the various datasets--Red raster: Device #1 Retrospective dataset, green raster: Device #1 Prospective dataset, blue raster: Alternate Device #2 dataset. (Panel H) Probability density histogram of the lead times of the RNN model (**not recalibrated**) predictions for each alert. Colors correspond to those in the raster plot. (Panel I) Raster plot visualization of timing of all true positive predictions (**not recalibrated**) relative to hard outcomes (Time 0). (Panel J) Probability density histogram of the lead times of the RNN model (**not recalibrated**) predictions for each hard outcome.

For the lead time analysis, we've kept the initial RNN model outputs and added a note to the caption that these results are from the RNN model that is not recalibrated. We did not reperform the lead time analysis since there were minimal changes in the performance metrics between calibrated and recalibrated models (as evident from figure 3 and the red lined updated table), apart from the changes in threshold, and thus the lead times should not change significantly.

Finally, we have also added the following line to the Methods, indicating this additional curve (Lines 285 – 286):

All RNN performance metrics were recalculated using the recalibrated model (Figure. 3, Panels A-F)

We wish to thank the reviewer again for this fourth round of thoughtful comments and suggestions. As you will see in this document, as well as the revised manuscript, the specific points raised by the reviewer have all been addressed; clarifications, table edits and figure updates have been added. We believe all this additional work has significantly improved the paper and we hope the revisions are satisfactory to the reviewer and the editors. Our responses are highlighted in blue and the corresponding additions/changes to the revised manuscript are highlighted in red. All line numbers refer to the red-lined manuscript.

Reviewer #3's comments:

The article is much improved by the changes made. I do not want to make significant additional work given the sequence of reviews that have been gone through. However, for the authors to consider.

- We'd like to thank the reviewer for the positive feedback—their comments and investment thus far have significantly improved the intellectual rigor of this paper.

1. I think it would be helpful to have flow diagrams to show how patients and data were included and excluded so that it is clear for example how there were 17 patients who deteriorated in table 1 but only 11 who are included in the assessment of performance.

- We'd like to thank the reviewer for pointing out the ambiguity in using the word "Deteriorated" in table 1. The 17 patients who "deteriorated" were actually those whose MEWS score reached a score of **greater than six** for more than 30 minutes, i.e. they were patients for whom we deemed had a valid and clinically meaningful **clinical alert**. This set of events were separate and distinct from the 11 hard outcome events of Table 3. To clear up this ambiguity we have modified third column of Table 1 to read "Clinical Alerts" instead of "Deterioration" (Lines XX – XX):

	Total			Retrospective	Alternate Device	Prospective
	All	Stable	Clinical Alerts	All	All	All
Patients (n)	888	871	17	315	413	160
Devices (n)	1302	1244	58	496	576	230
Male (%)	453 (51.2)	445 (51.3)	8.0 (53.3)	162 (51.4)	206 (49.9)	85 (53.1)
Race/Ethnicities, n (%)						
White	648 (73.3)	634 (73.1)	14.0 (93.3)	214 (67.9)	328 (79.4)	106 (66.2)
Multiracial	94 (10.6)	92 (10.6)	2.0 (13.3)	34 (10.8)	36 (8.7)	24 (15.0)
Black	86 (9.7)	85 (9.8)	1.0 (6.7)	32 (10.2)	30 (7.3)	24 (15.0)
Asian	38 (4.3)	38 (4.4)	0 (0.0)	26 (8.3)	9 (2.2)	3 (1.9)
Unknown/Other	16 (2.1)	18 (2.1)	0 (0.0)	9 (2.8)	6 (1.5)	6 (2.8)
Declined	4 (0.5)	4.0 (0.5)	0 (0.0)	0 (0.0)	4 (1.0)	0 (0.0)
Model Features, median [IQR]						
Age [median IQR]	69.3 [57.6, 78.0]	69.3 (57.5 - 77.3)	88.3 (86.5 - 90.5)	68.0 [56.0, 78.0]	70.0 [60.0, 79.0]	70.0 [56.8, 77.0]
BMI [median IQR]	27.0 (23.5 - 32.3)	27.4 (23.5 - 32.3)	30.4 (27.3 - 31.8)	27.4 [23.7, 32.1]	28.3 [23.9, 33.3]	26.6 [23.0, 31.5]
SMA [median IQR]	16.3 [8.9 - 17.7]	16.3 [8.9 - 17.7]	15.6 [9.1 - 16.3]	16.3 [8.9 - 17.7]	0.0 [0.0 - 0.0]	0.0 [0.0 - 0.0]
sBP [median IQR]	126.0 [109.3 - 137.7]	125.8 [109.3 - 137.7]	134.8 [120.3 - 147.8]	125.2 [110.0 - 138.0]	127.3 [112.0 - 140.0]	125.5 [106.0 - 135.0]
HR [median IQR]	79.8 [68.2 - 90.8]	79.8 [68.1 - 90.8]	79.8 [70.5 - 90.1]	79.7 [67.3 - 90.7]	75.7 [65.4 - 84.9]	84.0 [71.8 - 96.7]
RR (mean SD)	19.0 [16.5 - 21.3]	19.0 [16.5 - 21.2]	19.3 [17.3 - 21.2]	21.0 [17.4 - 23.9]	17.6 [16.0 - 19.1]	18.5 [16.1 - 20.8]
Temp (mean SD)	36.7 [36.3 - 36.8]	36.7 [36.3 - 36.8]	36.7 [36.3 - 37.1]	36.7 [36.3 - 37.1]	36.6 [36.4 - 36.7]	36.5 [36.1 - 36.7]
SpO2 (mean SD)	96.0 [95.0 - 97.9]	96.5 [95.3 - 98.2]	93.3 [91.1 - 96.3]	93.0 [90.8 - 96.0]	96.7 [95.4 - 99.0]	0.0 [0.0 - 0.0]
MEWS (mean SD)	3.0 [2.0 - 4.0]	2.9 [2.0 - 3.9]	5.6 [5.2 - 6.0]	3.1 [2.2 - 4.0]	3.0 [2.0 - 4.0]	2.8 [1.8 - 4.0]

I think it would be useful to the reader to qualify what the acceptable deviation lines for each vital were - so perhaps put in brackets next to 5/10% what these were - for temperature +/- 1- 1.5 degree C across the relevant range - for example.

- Thank you for highlighting out this point for additional clarity. To address this we've made the following edits to Lines XX – XX:

We also calculated the percentage of measurements that fell within the 5% or 10% error line. In subsequent analyses, we only used records where at least 2/3 (i.e., 67%) of a value for a given patient were within a 10% error for HR (+/- 3 - 20 bpm) and a 5% error for temperature (+/- 1.4 – 2 C) and SpO2 (+/- 4.25 – 5 %) from the EHR values.